EXP.NO: 4	Create a web form that accepts a user's name and age. Write a Servlet to process the form data and display it back on the browser.
DATE:26/02/2025	

AIM:

To create a simple web application that includes a form to accept a user's name and age, and then use a Servlet to process the submitted data. The Servlet will retrieve the form data, process it, and display the values back on the browser.

ALGORITHM:

1. Input:

The user fills out a web form with their name and age and submits it.

2. Form Handling and Servlet Processing:

1. Create the Web Form:

o Create an HTML form with two input fields: one for the user's name and one for the user's age.

o Add a submit button to the form.

o Set the forms action attribute to the URL of the Servlet and the method to POST (or GET).

2. Create the Servlet Class:

o Write a Servlet that processes the form data submitted by the user.

o The Servlet will override the doPost() (or doGet() depending on the form method) to handle form submissions.

o The Servlet will retrieve the name and age parameters from the request object using request.getParameter("name") and request.getParameter("age").

3. Process the Data in the Servlet:

o Extract the name and age values submitted from the form.

o Validate the input (e.g., check if the age is a valid number).

o Prepare a response that will display the user's name and age back on the browser.

4. Send Response to the Browser:

o The Servlet will send a response to the browser containing a message like:

"Hello, [name]! You are [age] years old"

- The response will be displayed to the user in the browser.

5. Deploy the Servlet:

- Compile the Servlet and deploy it to a Servlet container (like Apache Tomcat).
- Ensure the web.xml file or Servlet annotations are used to map the Servlet to the URL defined in the forms action attribute.

6. Test the Application:

- Open the web browser, fill in the form with the name and age, and submit it.
- The Servlet will process the data and display the message back to the user in the browser.

CODE:

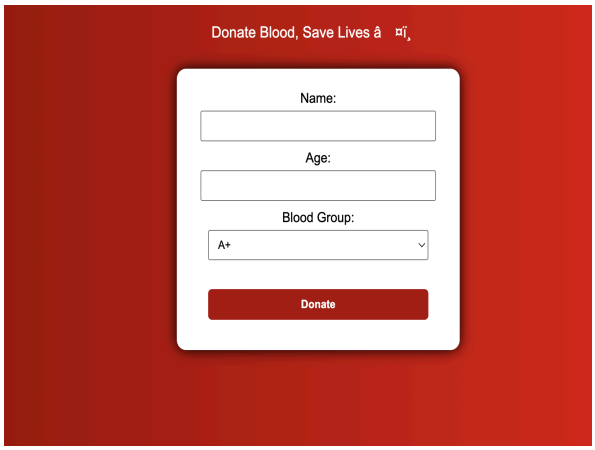
HTML FILE(index.html):

```
<!DOCTYPE html>
<html>
<head>
  <title>Blood Donation Form</title>
  <style>
     body {
        background: linear-gradient(to right, #8b0000, #ff0000);
        font-family: Arial, sans-serif;
        text-align: center;
        color: white;
     }
     h1 {
        margin-top: 30px;
     }
     p {
        margin-bottom: 30px;
        font-size: 18px;
     }
     form {
        background-color: white;
```

```
        color: black;
        padding: 25px;
        margin: auto;
        width: 320px;
        border-radius: 12px;
        box-shadow: 0px 0px 15px black;
    }
    input, select {
        width: 90%;
        padding: 8px;
        margin-top: 5px;
        margin-bottom: 10px;
    }
    input[type="submit"] {
        background-color: #b30000;
        color: white;
        border: none;
        padding: 10px;
        cursor: pointer;
        font-weight: bold;
        border-radius: 5px;
    }
    input[type="submit"]:hover {
        background-color: #800000;
    }
  </style>
</head>
<body>
<h1>🩸 Blood Donation Form</h1>
<p>Donate Blood, Save Lives ❤️</p>
<form action="blood4" method="get">
  Name:<br>
  <input type="text" name="name" required><br>
  Age:<br>
  <input type="number" name="age" required><br>
  Blood Group:<br>
  <select name="group">
    <option>A+</option>
    <option>A-</option>
    <option>B+</option>
```

```
      <option>B-</option>
      <option>O+</option>
      <option>O-</option>
      <option>AB+</option>
      <option>AB-</option>
   </select><br><br>
   <input type="submit" value="Donate">
</form>
</body>
</html>
```

OUTPUT:



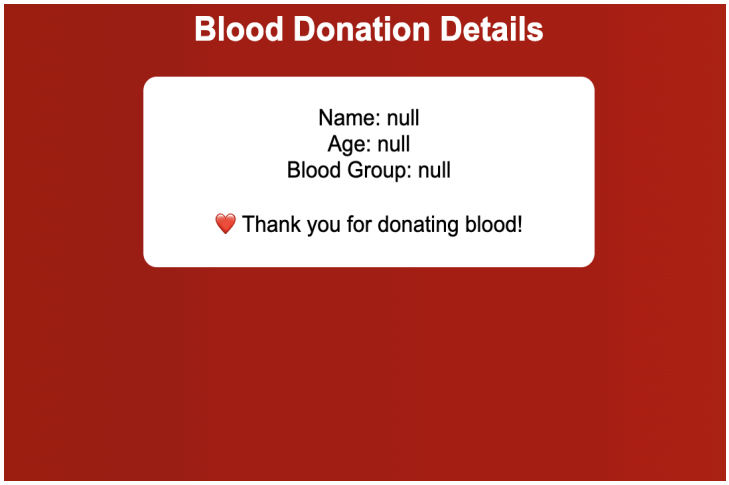
SERVLET:

```
package WTMA_exp;
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.*;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
@WebServlet("/blood4")
public class Exp4 extends HttpServlet {
   protected void doGet(HttpServletRequest req, HttpServletResponse res)
         throws ServletException, IOException {
      res.setContentType("text/html");
      PrintWriter out = res.getWriter();
      String name = req.getParameter("name");
```

```
        String age = req.getParameter("age");
        String group = req.getParameter("group");
        out.println("<html><head><style>");
out.println("body{background:linear-gradient(to
right,#990000,#cc0000);color:white;text-align:center;font-family:Arial;}");

out.println("div{background:white;color:black;padding:20px;margin:auto;width:300px;border-ra
dius:10px;}");
        out.println("</style></head><body>");
        out.println("<h2>Blood Donation Details</h2>");
        out.println("<div>");
        out.println("Name: " + name + "<br>");
        out.println("Age: " + age + "<br>");
        out.println("Blood Group: " + group + "<br><br>");
        out.println("❤️ Thank you for donating blood!");
        out.println("</div>");
        out.println("</body></html>");
    }
}
```

OUTPUT:



RESULT:

The Blood Donation web application was successfully developed using HTML and Servlet. The user input was processed, and the details were displayed correctly on the browser.